DEPARTMENT OF MATHEMATICS, IIT MADRAS

**MA5950 — Mathematical Finance: Theory & Implementation**

Instructor: Prof. Neelesh Upadhye

# Lecture 27: Assumptions and Limitations

What Black–Scholes Requires · What Markets Do · The Volatility Smile

October 11, 2026

# Today's Agenda

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4. Problem 2: hedging cannot be continuous
5. Problem 3: the volatility smile

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No physicist believes surfaces are frictionless. Yet the model isolates the essential mechanics, gives exact answers, and provides the baseline for adding friction later. Black–Scholes plays the same role. Nobody believes volatility is constant – but the entire derivatives industry is organized around *quoting deviations from this model*, which is far more useful than replacing it.

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*These adjust an input. The next two are genuinely difficult.*

# Problem 1: Returns Are Not Normal

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Under the normal model, each should not have occurred since the last ice age.

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# Problem 2: Hedging Cannot Be Continuous

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**Unbiased** – the model prices correctly on average.
But quarterly rebalancing leaves sd 3.2 on an option worth 10.45. Error $\sim 1/\sqrt{n}$.

# The Real-World Tension

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Rebalance more often $\Rightarrow$ less hedging error. Rebalance more often $\Rightarrow$ more transaction costs – which the model assumes away entirely. Every desk must choose a frequency trading these off, and Black–Scholes offers no guidance on it.

# Problem 3: The Volatility Smile

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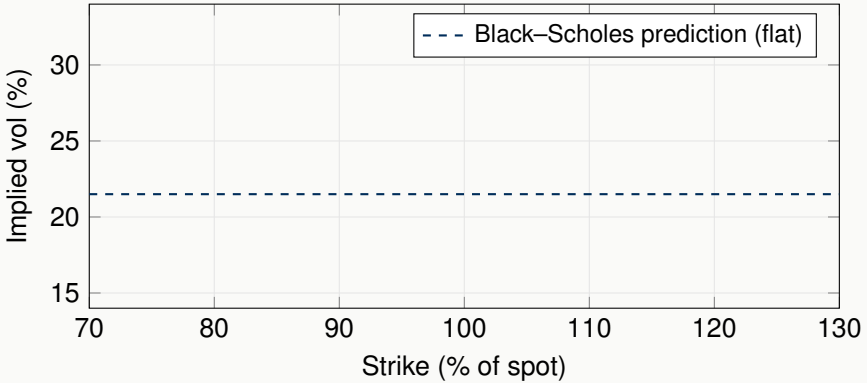
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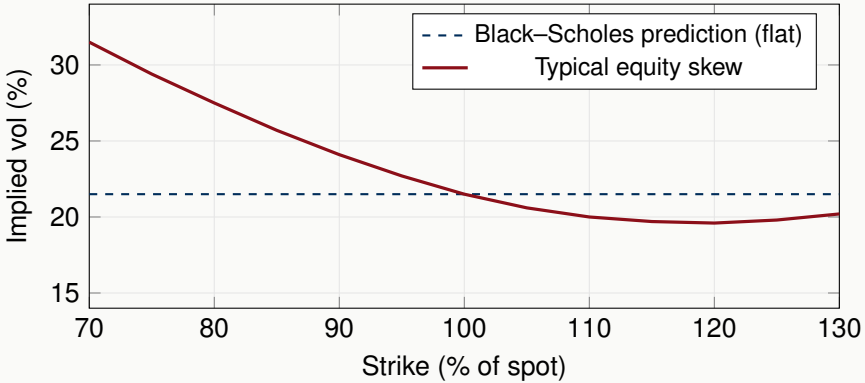
**Key Takeaway**

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# What Markets Actually Show



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*Illustrative.* Black–Scholes predicts a horizontal line.

# A Ruler That Reads Differently Each Time

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*The smile is Problem 1, expressed in the market's own prices.*

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What fails is one modelling choice: constant $\sigma$ and normal returns.

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- The **smile** falsifies constant $\sigma$ – yet the formula endures as a quoting language.

# Check Your Understanding

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3. Implied vol 28% at $K=90$, 19% at $K=110$. What does the market believe?

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Delta, Gamma, Vega, Theta, Rho – and running a hedged book.